

MID TERM EXAMINATION**Sep-2025****COURSE CODE__AFI507/RICV505****COURSE TITLE__Artificial Neural Networks****Time: 1:30 Hours****Max. Marks: 25****Note:** All questions are compulsory.
Assume suitable missing data, if any.

- Q.1 a) What are biological neurons? Draw the structure of biological neurons. How biological neurons help in creating artificial neural network?.
- b) Which learning is based on the concept of “cells that fire together, wire together”? Explain along with the weight update equation, applications and limitations.

[2+3 M][CO1][BTL#1,2]

- Q2. a) Consider the Boolean function f shown in table below, can you represent f with a single neuron neural network (i.e. no hidden layer)? If yes, design the network with weights. If not, justify the reason.

a	b	$f(a,b)$
1	1	0
0	0	0
1	0	1
0	1	0

- b) Explain ReLU and Leaky ReLU activation function with their derivatives.

[3+2 M][CO1][BTL#2]

Q.3 a) Consider a multilayer perceptron (MLP) model with 17 neurons in the input layer, 25 neurons in the hidden layer and 10 neurons in the output layer. What is the size of the weight matrix between the hidden layer and output layer, considering the bias to every neuron of output layer? Compute the total number of parameters of this model with bias and without bias.

b) Consider a network with weight sharing as shown in Figure 1. Let the activation function at hidden layer be $h()$ and for the output layer, be identity function. Compute the derivative of y w.r.t. $w_1^{(1)}$ during backpropagation. Show the derivation clearly step by step.

[2+3M][CO2][BTL#4]

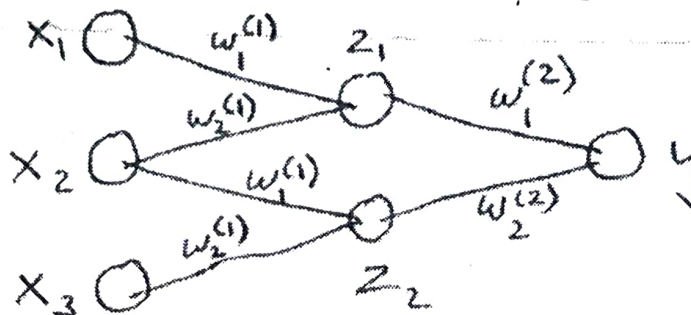


Figure 1

Q.4 a) Explain He initialization and Xavier initialization techniques with their advantages.

b) Differentiate between Batch gradient descent and Stochastic gradient descent with advantages and limitations.

[2.5+2.5 M][CO4][BTL#2]

Q.5 Consider a feed-forward neural network with one hidden layer shown in Figure 2 below:

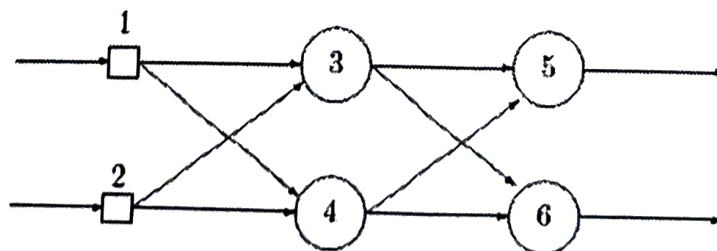


Figure 2. Neural Network

A weight on connection between nodes i and j is denoted by w_{ij} , such as w_{13} is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

$w_{13} = -2$	$w_{35} = 1$
$w_{23} = 3$	$w_{45} = -1$
$w_{14} = 4$	$w_{36} = -1$
$w_{24} = -1$	$w_{46} = 1$

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where v denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Derive the output equations y_5 and y_6 at nodes 5 and 6 respectively. Calculate the output of the network (y_5 and y_6) for each of the input patterns:

Pattern:	P_1	P_2	P_3	P_4
Node 1:	0	1	0	1
Node 2:	0	0	1	1

[5M][CO2][BTL#5]

Note: All questions carry equal marks.
Assume suitable missing data, if any.

Q.1 a) Consider a model that detects chest cancer using X-ray images. Which of the following is the most appropriate evaluation metric for such model: Accuracy, Precision, Recall, Loss. Justify your choice.

[3 M][CO 5][BTL 5]

b) Consider two data sets of similar size for a binary classification task. The first dataset contains almost entirely positive examples, and the second dataset contains only negative examples. You are training your model using minibatch gradient descent where you first train your model on mini-batches taken from first dataset followed by training on mini-batches taken from second dataset. Discuss the performance of your model.

[3 M][CO 5][BTL 5]

c) Consider a convolutional neural network (CNN) where only 5×5 sized kernels can be used, what is the minimum number of layers you would need to reduce the feature map size by a factor of two, given an input image of 128×128 ? You are not permitted to use any pooling layers, however, you may alter any other hyperparameters apart from the kernel size.

[4 M][CO 6][BTL 5]

Q.2 a) Why Convolutional Neural Networks instead of fully-connected networks are recommended for image recognition tasks. Justify the reasons.

[2 M][CO 4][BTL 6]

b) Consider a CNN classifier as shown in Figure 2. For each layer, calculate the number of parameters and the shape of output volume. The notation follows the convention: CONV-K-N denotes a convolutional layer with N filters, each filter of size K×K, padding and stride are 1 and 1 respectively. POOL-M indicates a M×M max pooling layer with stride M and padding 0. FLATTEN is the normal flatten layer. FC-N stands for a fully-connected layer with N neurons.

[8 M][CO 2][BTL 4]

Layer	Activation Volume Dimensions	Number of parameters
Input	$32 \times 32 \times 3$	0
CONV3-8		
Leaky ReLU		
POOL-2		
BATCHNORM		
CONV3-16		
Leaky ReLU		
POOL-2		
FLATTEN		
FC-10		

Figure 1

Q.3 a) List the key advantages of using pooling layers in a neural network? What differences would you expect between using a max pooling layer in comparison to an average pooling layer?

[5 M][CO 2][BTL 1,2]

b) Consider the following implementation of a convolutional neural network implementation shown in Fig. 2. The code is showing an error due to mismatching dimensions when performing the dot product between two matrices. The input image passed to the code is a grayscale image of dimension $32 \times 32 \times 1$. Given the error obtained, fix the code provided below.

[5 M][CO 3][BTL 6]

```

nn.Sequential(
  nn.Conv2d(in_channels=1, out_channels=8, kernel_size=5,
            stride=1, padding=0),
  nn.AvgPool2d(kernel_size=4, stride=4, padding=0),
  nn.ReLU(),
  nn.Conv2d(in_channels=8, out_channels=16, kernel_size=4,
            stride=1, padding=1),
  nn.AvgPool2d(kernel_size=2, stride=2, padding=0),
  nn.ReLU(),
  nn.Flatten(),
  nn.Linear(120, 10)
)

```

Figure 2

Q.4 a) Explain Back Propagation Through Time (BPTT) in Recurrent Neural Networks (RNNs) in details.

[5 M][CO 1][BTL 2]

b) Illustrate the problem of Exploding and Vanishing gradient in RNNs?

[5 M][CO 2][BTL 3]

Q.5 Explain any two of the following:

- i) Deconvolution
- ii) Residual network
- iii) Network Pruning Techniques
- iv) Boltzmann Learning

[5*2=10 M][CO 2][BTL 2]